INCH-POUND

MIL-R-25170B
w/AMENDMENT 1
6 September 2007
SUPERSEDING
MIL-R-25170B(USAF)
24 March 1983
And NOTICE 2
28 November 1997

MILITARY SPECIFICATION

RELEASE, AUTOMATIC, AIRCRAFT SAFETY, LAP BELT

Inactive for new design after 28 November 1997.

This specification is approved for use by the Department of the Air Force
and is available for use by all Departments and Agencies
of the Department of Defense

1. SCOPE

1.1 Scope.

This specification establishes the requirements for a gas-pressure-operated automatic-release mechanism for use with the automatic-opening safety lap belt and shoulder harness.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to ASC/ENOI, 2530 Loop Road West, Wright-Patterson AFB OH 45433-7101 or emailed to Engineering.Standards@wpafb.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1680

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2.2 Government documents.

2.2.1 Standards.

The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Mothods
MIL-STD-1916	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents.

The following drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these drawings are those cited in the solicitation or contract.

DRAWINGS U.S. AIR FORCE

55C502	Release, Automatic, Aircraft Safety Lap Belt
55B600	Hose-Automatic Lap Belt, Actuating, Assembly of

(Copies of these drawings should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence.

In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article test.

This specification makes provisions for first article testing.

3.2 Design and construction.

The automatic releases shall be constructed in accordance with Air Force Drawing 55C502.

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3.3 Government-furnished property.

When provided for in the contract or purchase order, the following items will be furnished by the Government to the contractor upon his request as required:

Initiator, Delay, M4, Part No. 94-3-7 or Initiator, Delay, M12, Part No. 94-3-9.

Cartridge, Delay, M46, Part No. BLX-93-1 or Cartridge, Delay, M71, Part No. F8493162.

The quantities of Government-furnished property are as follows: One delay initiator, loaded and staked, and two additional delay cartridges required per lot of automatic releases submitted for acceptance. See 6.3.

3.4 Identification of product.

The physical item shall be marked for identification in accordance with MIL-STD-130.

3.5 Workmanship.

Because of the emergency nature of the equipment, it is required that all details of workmanship be of the highest quality to insure proper functioning under the conditions to which the safety belts may be subjected. Unsatisfactory workmanship items such as loose, cocked, or inadequately headed rivets, distorted, loose, or inadequately secured bushings and pins, loose, broken, or incomplete stitching; and rough, malformed, misaligned, or improperly machined fittings shall be considered as defects. All parts shall be clean and free from dirt, sand, metal chips, or foreign matter while being assembled.

4. VERIFICATIONS

4.1 Responsibility for inspection.

Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.

The inspection and testing of releases shall be classified as follows:

a. Acceptance tests

b. First article tests

4.2.1 Acceptance tests.

Acceptance tents shall consist of

a. Individual tests

b. Sampling plan and tests

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4.2.1.1 Individual tests.

Each automatic release shall be subjected to the following tests as described under 4.4, Test methods:

a. Examination of product

b. Functional tests

4.2.1.2 Sampling plan and tests.

4.2.1.2.1 Lot.

The lot definition, formation, and size shall be in accordance with MIL-STD-1916.

4.2.1.2.2 Sampling tests.

Two automatic releases shall be selected at random from each lot of 200 or fraction thereof on the order. One of these releases shall be utilized for each of the following tests.

a. Low temperature automatic operation

b. Ultimate load

4.3 Test conditions.

Unless otherwise specified, all tests required by this specification shall be made under the normal atmospheric conditions existing at the time and location of the tests.

4.4 Test methods.

4.4.1 Examination of product.

Each automatic release shall be inspected for conformance to this specification with respect to workmanship, materials, dimensions, and markings.

4.4.2 Environmental tests.

One automatic release shall be subjected to the following environmental tests of MIL-STD-810. At the conclusion of all environmental tests, an initiator and a 48-inch length of pressure transmission hose (P/N 55B600-48) shall be attached to the release. The following loads shall then be applied to the belt as shown on figure 1: load A, 1000 pounds, loads B and C, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

Test	Procedure	Time
Humidity	I	240 hours
Salt Spray	I	50 hours
Sand and Dust	I	12 hours

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4.4.3 High temperature automatic operation.

The automatic lap belt assembly with initiator and a 36-inch length of pressure transmission hose (P/N 55B600-36) attached shall be subjected to a temperature of +160˚F (+71˚C), for a period of at least three hours. While still at this temperature, the following loads shall be applied to the belt as shown in figure 1: load A, 1000 pounds; loads B and C, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

4.4.4 Low temperature automatic operation.

The automatic lap belt assembly with initiator and a 48-inch length of pressure transmission (P/N 55B600-48) attached shall be subjected to a temperature of -65˚F (-55˚C) for a period of at least 4 hours. While still at this temperature, the following loads shall be applied to the belt as shown in figure 1: load A, 1000 pounds; loads B and C, 0 pounds. The initiator shall then be actuated. The connecting link shall readily separate from the release.

4.4.5 Ultimate load.

The release assembly shall be mounted on a suitable wooden block, representing the curvature of the anatomy, as shown in figure 1, and installed on a suitable static load testing machine. The release shall then be subjected to the following ultimate loads simultaneously: load A, 2880 pounds; loads B and C, 1800 pounds. There shall be no structural or mechanical failure of the release. The loads on the release shall then be reduced to the following: load A, 100 pounds; loads B and C, 50 pounds. The release shall then be operated automatically. Satisfactory release shall result.

4.4.6 Functional test.

After the 2000-pound static load has been applied to each belt assembly, as required by the belt assembly drawings, the AN520-0-4 screw shall be removed and the belt assembly placed in a vertical position with the manual release hanging below the automatic release. Fifty to one hundred psi air pressure of dry filtered air or nitrogen shall be applied to the AN823-3 elbow fitting of the automatic release. The link and lower half of the belt assembly shall fall from the automatic release. Any release which requires more than 100 psi to actuate the piston shall be rejected.

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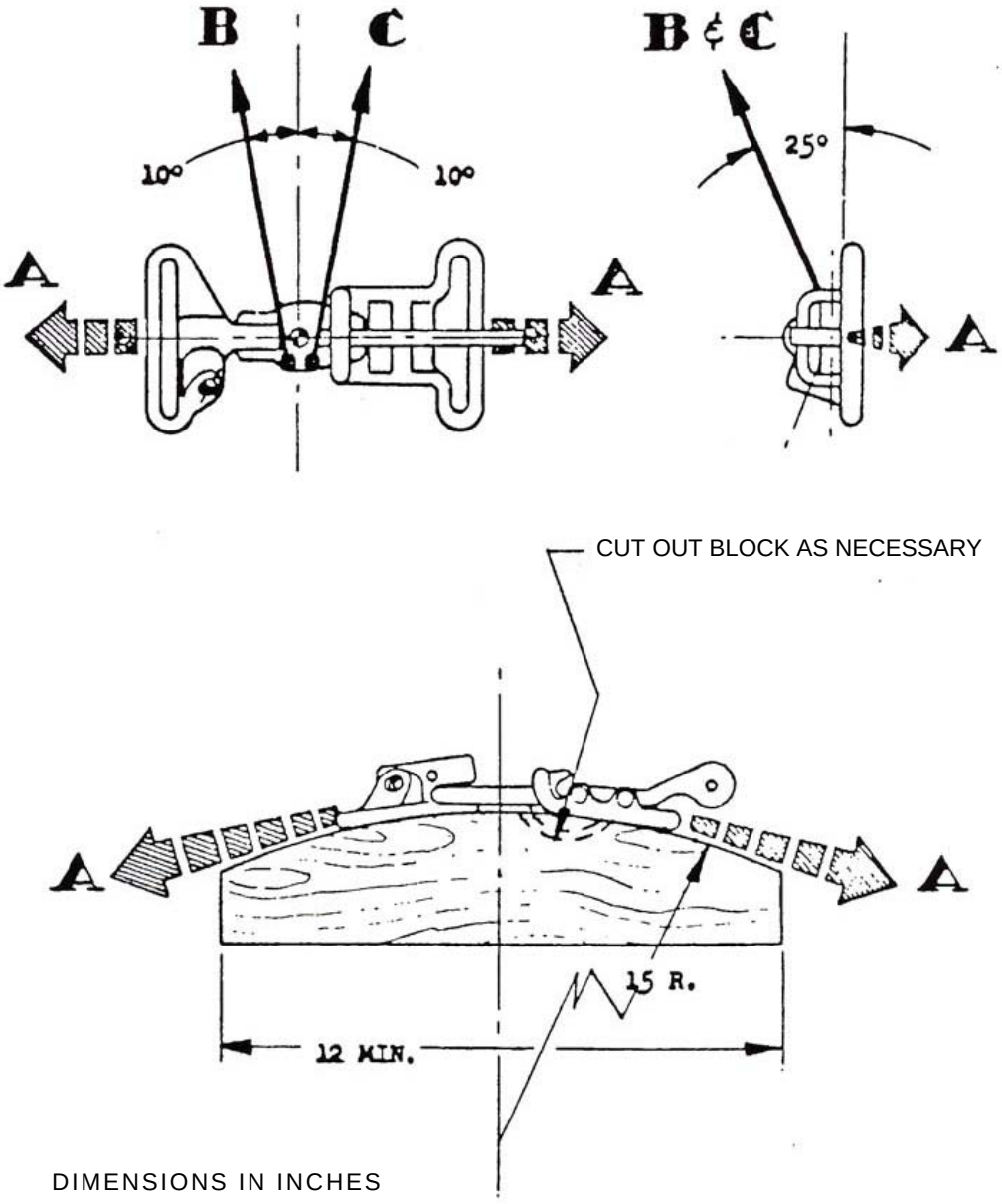


Figure 1. Load diagram.

4.5 First article tests.

4.5.1 First article test samples.

The first article test samples shall consist of models representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when

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so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 First article tests.

The first article tests of releases shall consist of all the tests specified in 4.4, Test methods.

5. PACKAGING

5.1 Packaging.

For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use.

The automatic release covered by this specification is compatible with the F-1A automatic parachute ripcord release anchor (P/N 55A751) and intended for use in USAF aircraft equipped with ejection seats in accordance with MIL-S-6551 and MIL-S-9479. (Reference Drawing 55D599.)

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Quantity of Government-furnished property (see 3.3).

c. Packaging requirements (see 5.1).

6.3 Reconditioning of initiators.

After firing, the delay initiator will be disassembled, cleaned, and reloaded by qualified armament personnel at AF depot level. The procuring activity will take the necessary action to coordinate arrangements for disassembly, cleaning, and reloading of these devices with the responsible depot. Upon completion of the initial reloading of the initiator, the body and cap should be painted solid red, a narrow black band painted around the body immediately below the cap, and marked "for ground test only." A second narrow black band should be added on

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the device after the second reloading. The proper nomenclature should be affixed to the device in accordance with the cartridge installed. These devices should not be fired more than three times, which permits only two reloadings on each unit. It must be noted that the components of each device should not be interchanged with those of another unit during disassembly, reloading, and assembly. Upon completion of the third firing of any initiator by the contractor (firing of a unit painted solid red with two black bands), the contractor should dispose of these units in accordance with instructions from the procuring activity.

6.4 Keyword list.

Restraint

Ejection

Seat

6.5 Changes from previous issue.

The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force – 11

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Air Force – 99
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(Project 1680-2007-002)

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